

ON-LINE SENSOR CALIBRATION MONITORING AND FAULT DETECTION FOR CHEMICAL PROCESSES

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ABSTRACT

In most process industries, periodic sensor calibrations are required to assure sensors are operating properly. Out-of-calibration sensors can result in decreased product quality, possibly causing a loss in revenue. Real-time, continuous sensor calibration monitoring is desirable to assure product quality and reduce maintenance costs associated with performing unnecessary manual sensor calibrations. An artificial neural network-based sensor calibration monitoring system can provide continuous sensor status information. This paper describes the design of a neural network-based instrument surveillance and calibration verification system (ISCV) for a chemical processing system.

INTRODUCTION

When monitoring complex processes, it is difficult, or impossible, to detect small drifts in sensor instrumentation. These drifts can cause incorrect control actions, poor product quality, and decreased process efficiency. The current method used to guard against calibration drifts is periodic sensor calibration. These calibrations usually require the instrument be taken out of service and be falsely-loaded to simulate actual in-service stimuli. This can lead to equipment damage and incorrect calibration due to adjustments made under non-service conditions. A less invasive technique is desired.

As increased economic competitiveness necessitates streamlining plant operations, condition based maintenance strategies rather than periodic or corrective maintenance strategies are desired. Changing calibration strategies to be condition-based requires that instruments be manually recalibrated only when their performance is degraded beyond a specific tolerance. Continuous verification of the instrument's calibration will reduce unnecessary sensor calibrations and give operators more confidence in sensor measurements. Elimination of unnecessary maintenance results in cost savings and reduced down time while a better knowledge of the actual state of the process, due to more reliable sensor values, could result in increased product quality, reduced equipment damage, and increased plant efficiency. Specifically, this system continuously monitors the condition of process sensors and allows for the automatic replacement of faulty sensor values with the system's best estimate. This system aids in scheduling maintenance and increases plant reliability.

SYSTEM OF INTEREST

Figure 1 is a simple block diagram of the chemical process of interest. The process has instrumentation which measure flows, temperatures, pressures and levels which need to be operating properly to ensure a high quality product. A neural network based sensor calibration and monitoring system can fulfill this need.

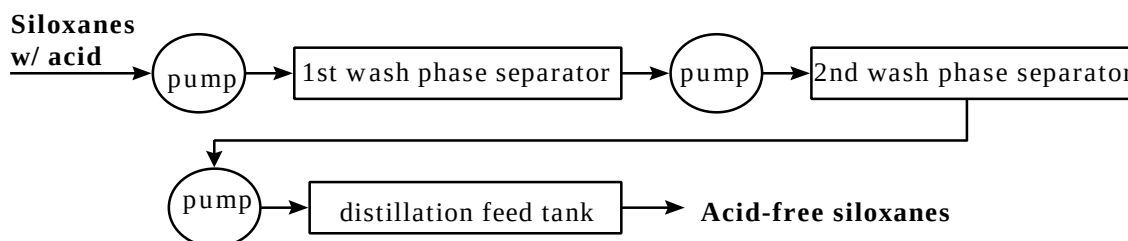


Figure 1. Siloxanes Wash/Separation Process

ISCV SYSTEM ARCHITECTURE

The artificial neural network (ANN) based instrument surveillance and calibration verification system (ISCV) has the following major components: a signal estimation module using autoassociative neural network (AANN) architecture, a statistical decision logic module based on the sequential probability ratio test (SPRT), a faulty sensor correction module, and a network tuning module. A block diagram of the ISCV system is shown in Figure 2.

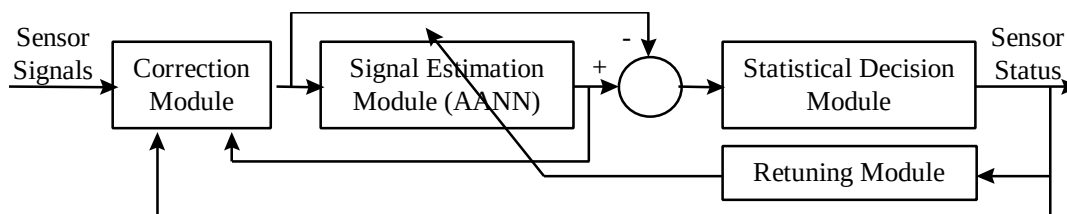


Figure 2. ISCV System Block Diagram

Signal Estimation Module

The use of AANNs for plant wide monitoring has been widely reported in the nuclear industry [B. R. Upadhyaya and E. Eryurek, 1992, R. E. Uhrig, et al, 1996, Nabeshima, et al, 1995]. Similar work using ANNs applied to chemical process systems have also been reported [Dong and McAvoy, 1994, Kramer, 1992]. The work presented in this paper advances the AANN methodology by introducing a faulty sensor replacement algorithm and a model tuning procedure. This research is also significant because it uses data from a chemical process which is not instrumented as fully as nuclear power plants studied previously [Hines 1998].

In an autoassociative neural network, the outputs are trained to emulate the inputs over an appropriate dynamic range. Many plant variables that have some degree of coherence with each other constitute the inputs. During training, the interrelationships among the variables are embedded in the neural network connection weights. A robust training procedure is used to force

the network to rely on redundant information from correlated sensors to estimate that specific sensor's value. As a result, any specific network output shows virtually no change when the corresponding input has been distorted by noise, faulty data, or missing data. This characteristic allows the AANN to detect sensor drifts or failures by comparing sensor measurements (network inputs) with the corresponding network estimates of the sensor values (network outputs).

Figure 3 shows a sensor monitoring module for a group of four sensors whose measurements are correlated to some degree (actual networks have 15-30 correlated sensors as inputs). When a sensor's signal to the autoassociative network is faulty due to a drift or gross failure, the network still gives a valid estimate of the sensor value due to its use of information from other correlated sensors. The difference or residual (r_n) between the sensor estimate (s_n') and the actual measurement (s_n) normally has a zero mean and a variance related to the amount of noise in the sensor's signal. When a sensor is faulty, its associated residual's mean or variance changes. This can be detected with the statistical decision logic.

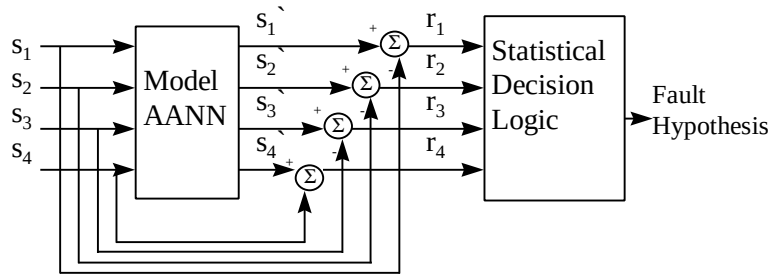


Fig. 3. Sensor Monitoring Module

Statistical Decision Logic Module

The decision logic module implements the SPRT which was initially developed by Wald [1945], later used by Upadhyaya [1992], and more recently by Gross and Singer [Gross 1991]. This module uses the residual as the input and outputs the condition of the sensor. The output of zero indicates sensor is normal, while an output of unity indicates the sensor is faulty. Rather than computing a new mean and variance at each sampling time, the SPRT continuously monitors the sensor's performance by adding the information contained in each residual to compute a likelihood of failure. This SPRT-based method is optimal in the sense that a minimum number of samples are required to detect a fault existing in the signal.

When a sensor is operating correctly, the residual should have a mean of zero, and a variance comparable to that of the sensor (due to the filtering characteristics of the AANN). If there is sensor drift, the residual mean shifts. Due to the shift in residual mean, the likelihood ratio increases. This ratio is a measure of how close the residual is to zero. If the likelihood ratio increases above a certain predefined boundary (user specified by false and missed alarm probabilities), the residuals are determined to be more likely from the faulted distribution than from the unfaulted distribution, and the sensor is classified as faulted. When the likelihood ratio is less than the boundary, the sensor is classified as normal. If a sensor is determined to be faulty, the likelihood ratio is reset to zero and the calculation to determine the status of the sensor begins again.

Faulty Sensor Correction Module

The statistical decision module continues to monitor a sensor output even after it has been determined to be faulty. While the sensor is faulted, the best estimate of the sensor value (the neural network output) can be used for input into control systems, for display to plant operators, or for other critical tasks. The best estimate also replaces the faulty sensor as input into the AANN so that the monitoring of other sensors is not degraded. The actual sensor output is substituted back into the network when the fault has been cleared. This method always gives the operator or control system access to the best estimate of the parameter whether it is the unfaulted measured value or the estimated value.

Network Tuning Module

The neural network learns the interrelationships among the sensors measurements during training. Although the training set should include samples from all plant operating regions, sometimes the operating state may change to one that was not included in the training set. This can be caused by component wear, cyclical changes, or changes in the plant configuration, among others. These changes would be characterized by several residuals deviating significantly from their normal mean of zero. When this happens, the output of the AANN is not reliable and the network must be retrained to operate under the new conditions. If only one residual deviates from zero, a sensor fault is hypothesized.

The AANN architecture used is a three hidden layer feedforward network proposed by Kramer [1992]. It consists of an input layer, 3 hidden layers (mapping layer, bottleneck layer and demapping layer) and an output layer. Kramer points out that this network structure is necessary to model non-linear processes and provides access to the non-linear principle component analysis which are the values of the bottleneck layer.

The hidden layers act as feature extractors and the linear output layer combines these features to provide a desired mapping. If the features do not change when a plant or process operating condition changes, only the output layer weights need to be adjusted to perform the desired mapping without retraining the entire network. This assumption seems to hold for small changes in operating conditions. Retraining the entire network may be necessary for major changes in plant operating conditions when adjusting the output weights does not result in satisfactory performance.

Retraining only the linear output layer can be achieved by solving for the output layer weights using a least squares procedure. Several methods of solving for the linear output weights exist including pseudo-inverse methods that can cause numerical inversion problems, better methods use the LU or QR decompositions. The best method uses the singular value decomposition (SVD) technique which uses the most relevant information to compute the weight matrix and discards unimportant information that may be due to noise.

RESULTS

This section presents the results of the ISCV system performance when sensors are corrupted with artificially induced drifts in individual signals. The signal estimation module and the statistical decision logic module work together to determine sensor status, the faulty sensor replacement module provides correct inputs to the network, and the retuning module adapts the network to new operating conditions.

Small Drift Detection

A drift error is defined as a slow rate of change in a signal's correct value. To test the performance of the networks, both high and low drift faults were artificially induced in each of the 19 network channels. Simulations were performed to see how soon the system could detect the fault with a minimum of false alarms. In both test cases, faults were initiated at time stamp 5000. Figure 4a shows an example of the separator sump level signal artificially drifting low (2% per day ramp drift). The top plot contains the actual signal (drifting) and the network's best estimate, the middle plot is of the residual, and the bottom plot displays the output of the SPRT decision logic module. At about time stamp 8000, the sensor value drops from its median value of 40% to 37.2%, with an associated change in the residual mean of approximately 2.8%. This exceeds the fault tolerance of the SPRT, and an alarm is initiated. Figure 4b shows the change in the weak acid recycle control signal with a step drift of 5%. The sensor signal began to drop at time stamp 5000 by 5%. This value also exceeds the fault tolerance of the SPRT, and the ISCV system detects the fault at about time stamp 5300.

The spurious alarms before the sensor fault occurs are due to excessive noise in the sensor signal, which do not indicate a sensor fault has occurred. These spurious alarms can be reduced through additional training or relaxing the drift detection threshold. Continuous alarms beginning at about time stamp 8000 indicate that the sensor is faulty.

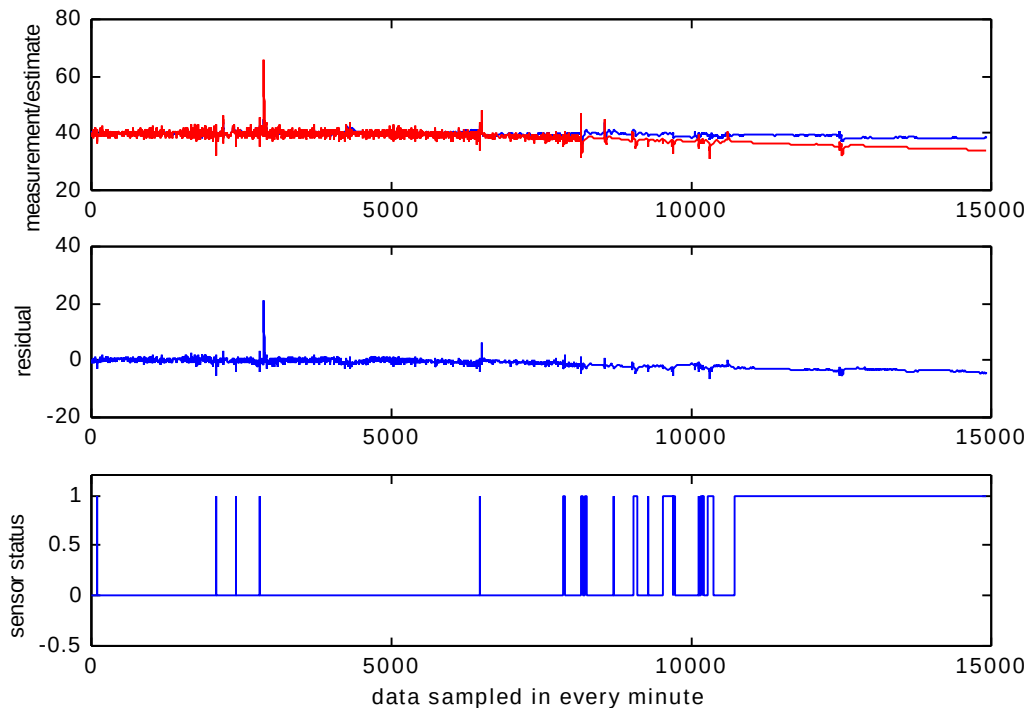


Figure 4a. Detection of a 2% per day ramp drift low in separator sump level

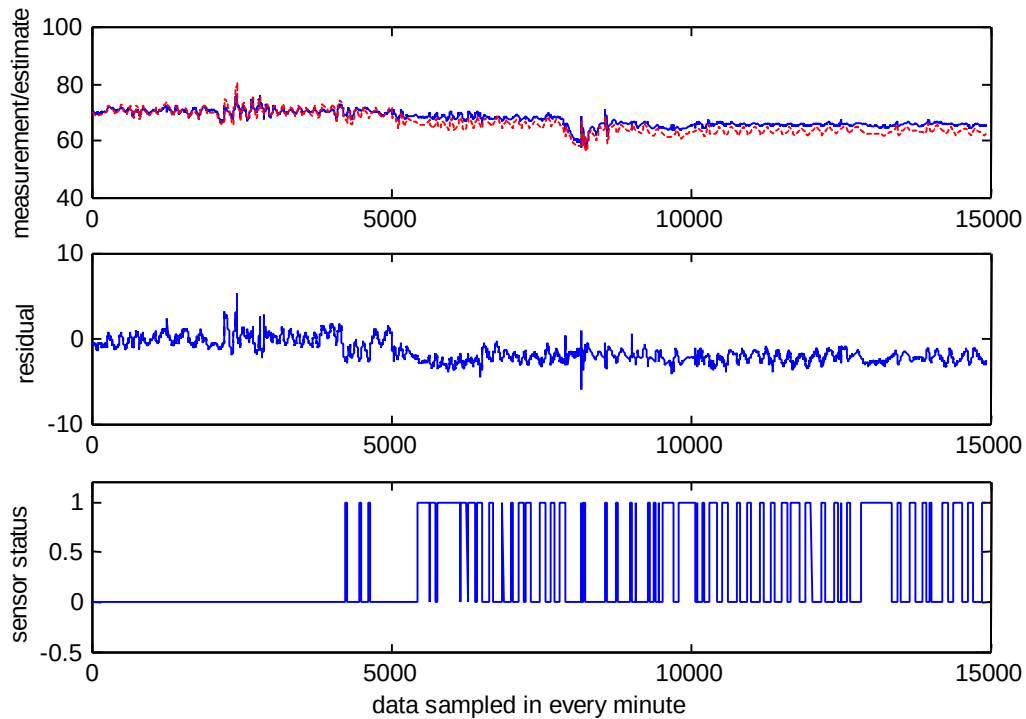


Figure 4b. Detection of a 5% step drift low in weak acid recycle control

Gross Fault Detection

Gross faults are defined in this study as drastic changes in the signal values. Gross faults are simulated by failing the sensor to its maximum or minimum full-scale deflection, representing gross fault "high" or gross fault "low", respectively.

Depending on how "gross" the signal fails, the network may or may not remain stable. A large drop in a signal's value may cause other parameters in the network to vary in an attempt to compensate for such a large loss of information. A larger fault can create false alarms in other channels due to the instability of the network. The residuals may change to a degree that they are greater than the pre-set faulted mean values of the SPRT's. While the other parameter residuals may vary, it is only a fraction of the amount that the signal that contains the gross fault varies.

Figure 5 shows a gross fault condition in which the separator sump level fails low at time stamp 5000. Due to the total loss of information, the ISCV system is unable to fully recover the original signal, but it is still able to recover the majority of the lost information of that signal (about 86%) and initiates a fault alarm right after the fault occurs. It should be pointed out that in a redundant system, more information could be recovered for a sudden information loss situation. Also, the faulty sensor replacement module will replace the input with the best estimate thus restoring the correct signal.

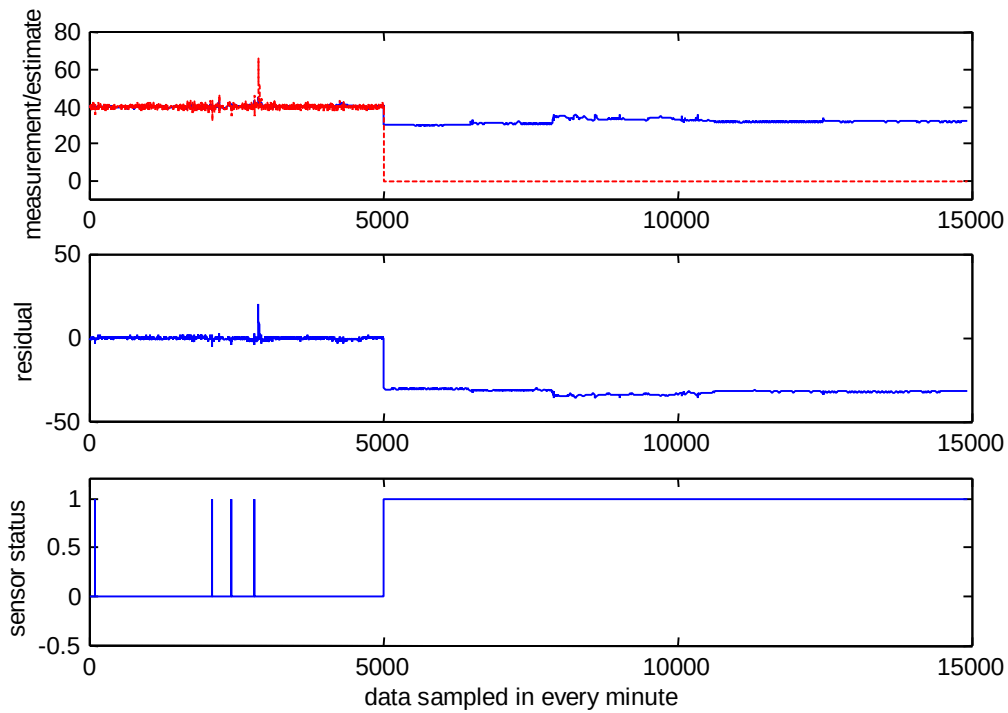


Figure 5. Detection of a gross fault low in separator sump level

Faulty Sensor Replacement

When a sensor is faulty, the fault may not be corrected immediately due to operational constraints. Thus, it is desirable for the system to output the correct value of the failed sensor so that the plant could continue operating without interruption. This is especially desirable when the faulty signal is a controlled variable in a feedback control system. In addition, this would minimize degradation of the ISCV system because the faulty input is replaced with a fault-free signal. The correction module designed in this study is capable of immediately replacing faulty sensor signals with their best estimates.

Figure 6 shows the corrected signal result when a 5% step drift was induced in the 2nd wash loop temperature signal. It indicates that when the drift is within the fault tolerance, the SPRT continues monitoring without any false alarms until the drift exceeds the tolerance at time stamp 200. Immediately after the fault tolerance of the SPRT is exceeded, the SPRT initiates an alarm and the correction module replaces the faulty signal with the system's best estimate. After the faulty signal is replaced, no further alarms occur (see bottom plot of Figure 6). The middle plot shows the residual between the measurement and the corrected signal.

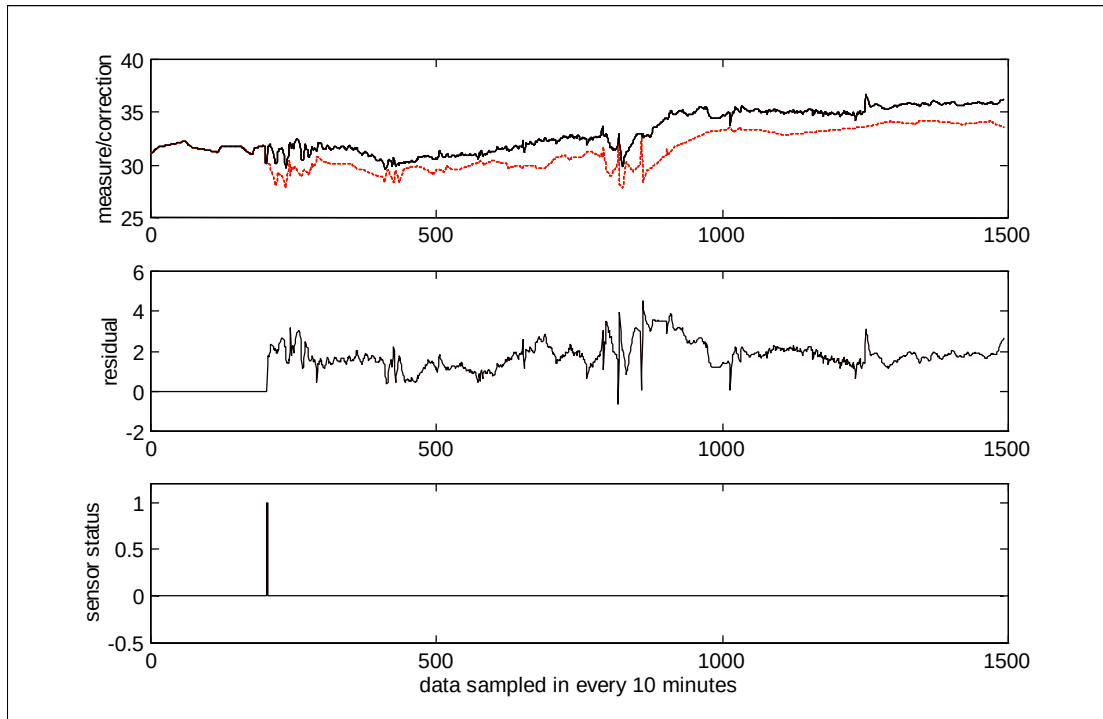


Figure 6. Corrected signal with a 5% step drift low in 2nd wash loop temp.

Model Changes and Adaptation

Neural networks are not guaranteed to function as expected when operating outside the training region. Although they have good generalization abilities inside the training space, they must be retrained when expected to operate in new regions. When operating conditions change slightly, it may not be necessary to completely retrain the network, but retune the output layer by using a linear regression technique: the Singular Value Decomposition (SVD), which is fast and can satisfy on-line adaptation requirements.

The original operation data were not found to have significant operation condition changes, therefore, artificial changes have been induced for system testing. The following results were obtained by simulating a 10% change in the operation beginning at time stamp 500. Figure 7 presents an example using the weak acid recycle control signal. The results show that the ISCV system adapted itself to the new operating conditions. No additional alarms resulted after retuning. The spurious alarms occur due to excessive noise and could be removed through filtering.

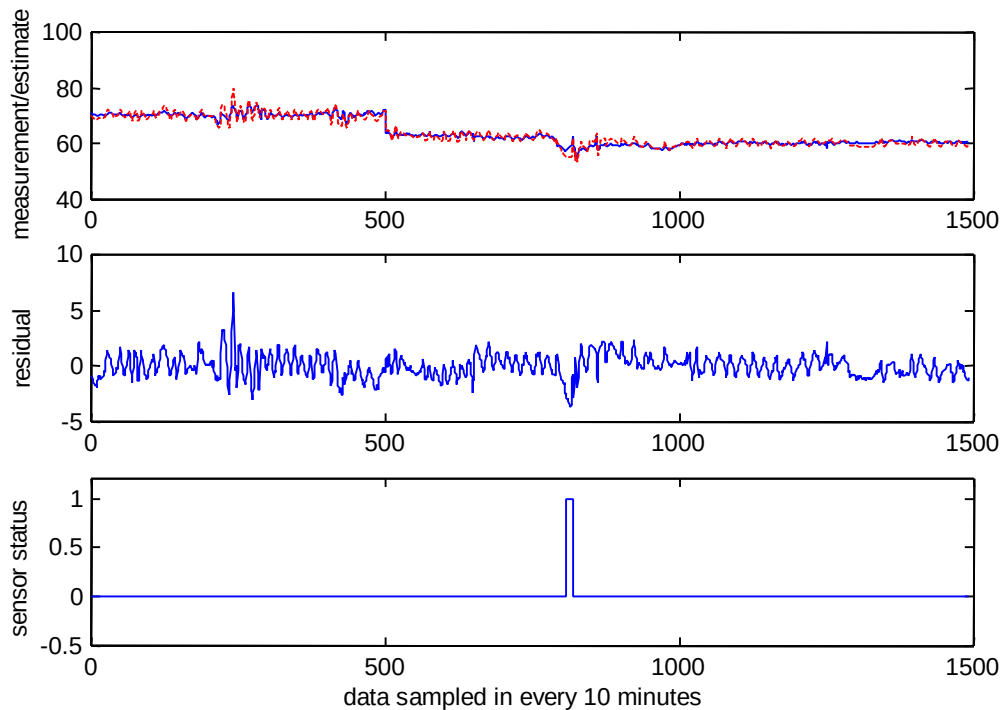


Figure 7. System adapting to new operation condition (weak acid recycle control)

CONCLUSIONS

It was found that due to the relatively small number of sensors to be monitored, using linear correlation coefficient analysis to refine the parameter selection was effective and simple. Although, high linear correlations between network parameters was not found to be a strict requirement for optimal system performance. The three hidden layer "feature detection" autoassociative neural network trained on a robust training set was shown to have excellent generalization abilities making it ideal for a plant wide sensor monitoring system such as the one implemented in this study.

The SPRT method has proven to be an excellent detection tool for incipient drift faults as well as gross faults. During operating condition changes, the SVD technique effectively retunes the network so that the ISCV system can quickly adapt itself to the new operation conditions without producing false alarms.

It was also found that the robustness of the monitoring network is related to the amount of signal redundancies and the degree of signal correlations. The system was able to detect faults at levels between 0.42-13% of the sensor's full-scale deflection. The level was dependent on the degree of correlation between signals and the amount of noise in the signals. The average detection level was about 3.2% of full-scale deflection, which is much higher than that in nuclear power plants (1% levels) which are more fully instrumented [Uhrig, Hines, Black, Wrest, Xu, 1996]. The ISCV system using an autoassociative neural networks can continuously monitor sensors for faulty operation.

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